

Product Requirements Document (PRD)

PharmaConnect – Pharmacy Inventory Intelligence & Redistribution Platform

1. Executive Summary

PharmaConnect is a digital platform designed to connect community pharmacies, hospital pharmacies, and other licensed healthcare facilities. The platform helps reduce medication wastage caused by expiry, improves inventory utilization, identifies dead stock, provides purchasing intelligence, and enables pharmacy-to-pharmacy redistribution of near-expiry medications.

2. Problem Statement

Pharmacies frequently experience inventory inefficiencies including expired medications, dead stock, poor demand forecasting, over-purchasing of slow-moving products, and difficulty locating urgently needed medications from nearby facilities.

3. Product Vision

To become the leading pharmacy inventory intelligence and redistribution network that minimizes medication waste, improves access to medicines, and increases pharmacy profitability through data-driven decision making.

4. Target Users

- Community Pharmacies
- Hospital Pharmacies
- Pharmacy Chains
- Pharmaceutical Procurement Teams
- Inventory Managers
- Licensed Healthcare Facilities

5. Core Features

A. Pharmacy-to-Pharmacy Redistribution

- List near-expiry medications.
- Search for urgently needed medications.
- Match buyers with sellers.
- Enable discounted transfers before expiry.

B. Inventory Tracking

- Real-time inventory visibility.
- Batch and expiry management.
- Stock movement tracking.

C. Expiry Alert System

- Alerts at 90, 60, and 30 days before expiry.
- Dashboard notifications.
- Suggested redistribution opportunities.

D. Dead Stock Detection

- Identify slow-moving inventory.
- Highlight products with low turnover.
- Recommend reduction of future purchases.

E. Demand Forecasting

- Analyze historical sales patterns.
- Predict future medication demand.
- Recommend reorder quantities and timing.

F. Profit Analysis

- Inventory profitability dashboard.
- Lost revenue analysis.
- Expiry loss tracking.

G. Smart Purchase Recommendations

- Recommend high-demand medications.
- Flag products with poor performance.
- Regional demand insights.

6. User Stories

- As a pharmacist, I want to receive expiry alerts so I can prevent inventory losses.
- As a pharmacy owner, I want to identify dead stock so I can improve profitability.
- As a hospital pharmacist, I want to locate medications from nearby pharmacies when urgently needed.
- As an inventory manager, I want demand forecasts so I can optimize purchasing decisions.

7. MVP Scope

Phase 1 should include:

- Pharmacy registration and verification.
- Inventory upload and management.
- Expiry alerts.
- Pharmacy-to-pharmacy medication listings.
- Dead stock identification.
- Basic analytics dashboard.

8. Future Enhancements

- AI-powered demand forecasting.
- Integration with pharmacy management systems.
- Automated inventory synchronization.
- Patient medication availability search.
- Supplier marketplace integration.
- National medication demand heatmaps.

9. Revenue Model

- Monthly subscription plans.
- Transaction fees on successful redistribution.
- Premium analytics packages.
- Enterprise plans for pharmacy chains and hospitals.

10. Success Metrics

- Reduction in expired inventory.
- Increased inventory turnover.
- Number of successful redistributions.
- Revenue saved from expiry prevention.
- Monthly active pharmacies.

11. Risks and Considerations

- Regulatory compliance.
- Verification of licensed facilities.
- Data privacy and security.
- Inventory data accuracy.
- Preventing off-platform transactions.

12. Conclusion

PharmaConnect addresses a significant challenge within the pharmacy ecosystem by combining inventory intelligence, expiry management, demand forecasting, and pharmacy-to-pharmacy redistribution into a single platform. The solution aims to reduce waste, improve medicine availability, and increase profitability across the healthcare supply chain.